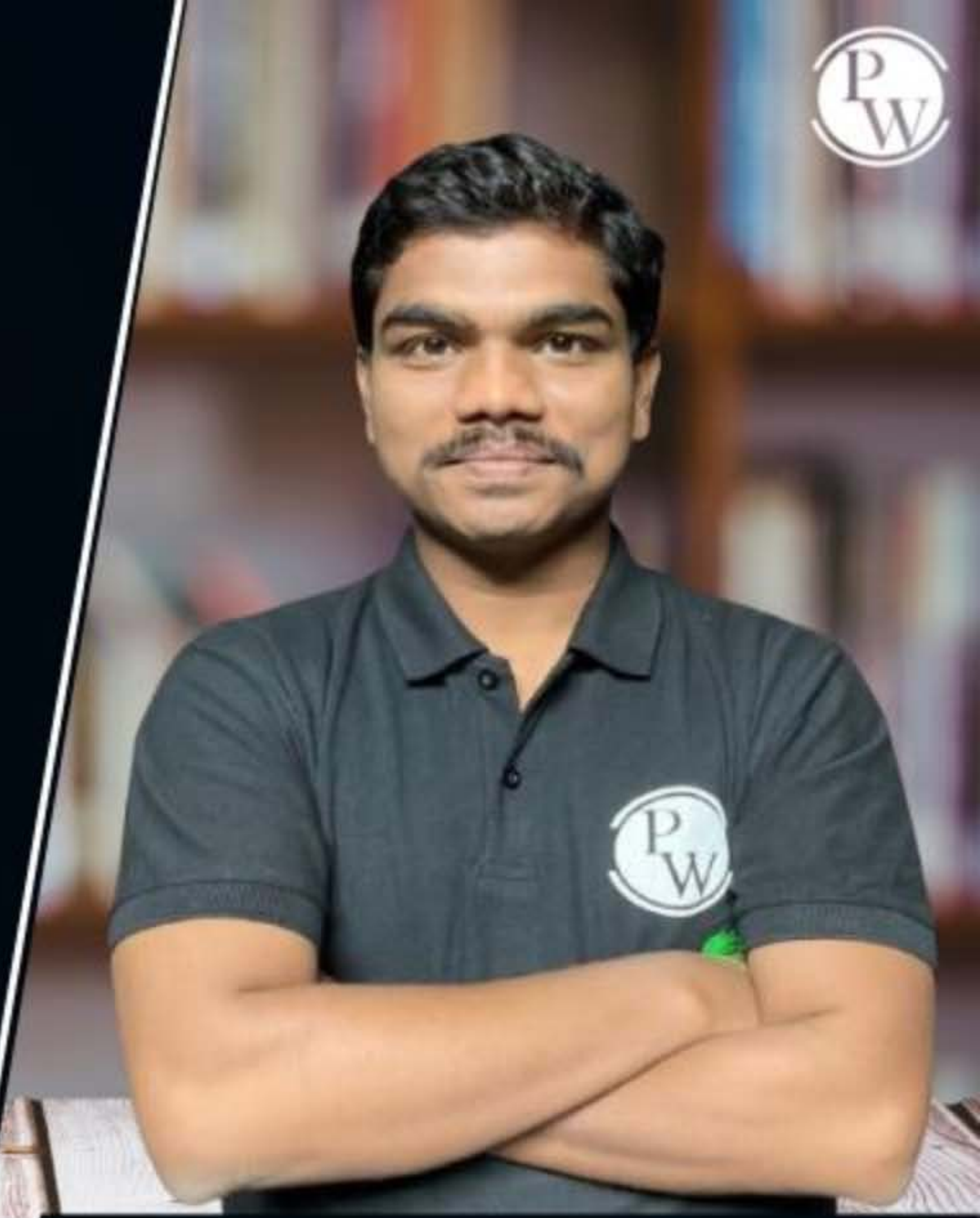
GATE- ALL BRANCHES
ENGINEERING MATHEMATICS

Linear Algebra

Lecture No. 1



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Recap of previous lecture

→ Complex Integrals

→ Cauchy's Residue Theorem;

$$\oint_C f(z)\,dz = 2\pi i \sum_{j=1}^{k} \operatorname{Res}(f(z))\Big|_{z=z_j}$$

→ Real definite Integrals.

Topics to be Covered

Topic Introduction to Linear Algebra

Topic Types of Matrices

Introduction to Linear Algebra

Operations of Matrices on vectors.

$$a_{11}x + a_{12}y = b_1$$
$$a_{21}x + a_{22}y = b_2$$

Easily Solvable.

$$\underbrace{A\vec{x} = \vec{y}}$$

→ A System with 30 Equations &

30 unknowns.

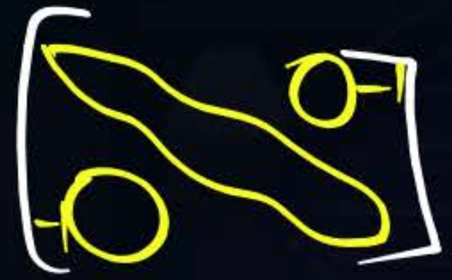
$$\boxed{\sum_{i=1}^{30}\sum_{j=1}^{30} a_{ij}x_j = (b_i)}$$

→ Signal Processing

$W = [\quad]$

→ Machine learning Algorithms.

→ Sparse systems



→ **Matrix:** A matrix is an array (arrangement) of elements in horizontal lines (called rows) & Vertical lines (columns).

A Matrix is denoted by Capital Letters.

Ex:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}_{3\times 3} \quad ; \quad B = \begin{bmatrix} v & i & n & a & y \\ l & o & v & e & s \\ i & n & d & i & a \end{bmatrix}_{3\times 5}$$

→ **Order of a Matrix:** A matrix 'A' is said to be of order $m\times n$ (read as 'm' by 'n') if the matrix has 'm' rows & 'n' columns.

→ If $m \neq n$; then the matrix 'A' ⟶ Rectangular Matrix.

If $m = n$; then the Matrix 'A' ⟶ Square Matrix.

→ Indicial Notation:

Matrix is of size $m \times n$.

A matrix $A = [a_{ij}]$ where $1 \leq i \leq m$; $1 \leq j \leq n$ is defined by

$a_{ij} = f(i, j)$.

a_{ij} ⟶ Element in i^{th} row & j^{th} column.

i → Denotes the row number.

j → Denotes column number.

Ex: If $A = [a_{ij}]$ where $1 \le i \le 2$; $1 \le j \le 3$ and $a_{ij} = i + j$

then $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}_{2\times 3}$

$a_{11} = 1 + 1 = 2$

$a_{12} = 1 + 2 = 3$

Ex: If $A = [a_{ij}]$ where $1 \le i \le 2$; $1 \le j \le 2$ where $a_{ij} = \int_i^j x^2\,dx$.

then $A = \begin{bmatrix} 0 & 7/3 \\ -7/3 & 0 \end{bmatrix}_{2\times 2}$

$a_{11} = \int_1^1 x^2\,dx = 0$

$a_{12} = \int_1^2 x^2\,dx = \left.\frac{x^3}{3}\right|_1^2 = \frac{8}{3} - \frac{1}{3} = 7/3$

$a_{21} = \int_2^1 x^2\,dx = -7/3$

$a_{22} = \int_2^2 x^2\,dx = 0$

→ **Principal diagonal of the Matrix:**

If $A = [a_{ij}]_{n\times n} \Rightarrow A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$ → Principal diagonal

then the diagonal that contains all elements a_{ii} is called the Principal diagonal / Leading diagonal of the Matrix.

→ **Trace of the Matrix: $tr(A)$:**

The sum of all the Principal diagonal elements of the matrix is called trace of the Matrix. i.e $tr(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}$.

Types of Matrices

① Upper Triangular Matrix:

A Matrix 'A' is said to be an upper Triangular Matrix if all the lower Triangular elements of the Matrix are zeroes.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

upper Triangular elements

lower Triangular elements.

$\Rightarrow a_{21} = a_{31} = a_{32} = 0$.

$\longrightarrow$ A matrix $A = [a_{ij}]$ where $1 \le i \le n$; $1 \le j \le n$ is said to be upper Triangular Matrix if

$$a_{ij} = 0 \;\forall\; i > j$$

Ex: $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}$

② Lower Triangular Matrix:

A Matrix 'A' $= [a_{ij}]$ where $1 \le i \le n$, $1 \le j \le n$ is a lower Triangular Matrix if $\underbrace{a_{ij} = 0 \;\forall\; i < j}$

i.e All upper Triangular elements are zero.

$$A = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

③ Diagonal Matrix:

A Matrix $A = [a_{ij}]$; $1 \le i \le n$; $1 \le j \le n$ is said to be a diagonal Matrix

if $a_{ij} = 0 \ \forall \ i \neq j$

$$A = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}.$$

This Matrix is also denoted as

$$A = \text{diag}[a_{11}, a_{22}, a_{33}]$$

④ Scalar Matrix: A Matrix 'A' $= [a_{ij}] \ \forall \ 1 \le i \le n, 1 \le j \le n$ is a Scalar Matrix

Where $a_{ij} = \begin{cases} K; & \forall \ i = j \\ 0; & \text{Otherwise.} \end{cases}$ & 'K' is a constant.

Ex: $A = \begin{bmatrix} K & 0 & 0 \\ 0 & K & 0 \\ 0 & 0 & K \end{bmatrix}$

When $K = 1$; $A \longrightarrow$ Identity Matrix; It is denoted by I → Multiplicative Identity.

$$I = [a_{ij}]_{n \times n} \text{ where } a_{ij} = \begin{cases} 1; & i = j \\ 0; & i \neq j \end{cases}$$

When $K = 0$; $A \longrightarrow$ Null or zero Matrix; It is denoted by O → Additive Identity.

$$O = [a_{ij}]_{n \times n} \text{ Where } a_{ij} = 0 \ \forall \ 1 \leq i \leq n, 1 \leq j \leq n.$$

Transpose of a Matrix:

A Matrix '$B_{n\times m}$' is Said to be transpose of a Matrix '$A_{m\times n}$' if

$\underbrace{a_{ij} = b_{ji}}$ $\forall$ $1 \le i \le m$; $1 \le j \le n$.

Ex: Say $\underbrace{A_{2\times 3} \;\&\; B_{3\times 2}}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}; \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

$\therefore$ Matrix B (i.e Transpose of 'A') Can be obtained by interchanging Rows & columns of 'A'.

It is denoted as $B = A^T = A'$.

Ex: If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3\times 2}$ then $B = A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}_{2\times 3}$

Properties of transpose:

① If 'A' is a square Matrix of size $n \times n$; then

a) $tr(A) = tr(A^T)$ (Since Principal diagonal elements don't change their Position by transposing a Matrix).

b) $\det(A) = \det(A^T)$

② $(A+B)^T = A^T + B^T$.

③ $(A-B)^T = A^T - B^T$.

④ If 'α' is a scalar, then

$$(\alpha A)^T = \alpha A^T.$$

Ex: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow 2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} \Rightarrow (2A)^T = 2\underbrace{\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}}_{A^T}$

→ If $A = [a_{ij}]_{m \times n}$ & $B = [b_{ij}]_{m \times n}$

then $\underset{m \times n}{C} = A + B$ is defined as

$$\boxed{c_{ij} = a_{ij} + b_{ij}}$$

$$c_{ji} = a_{ji} + b_{ji}$$

(5) Symmetric Matrix:

A Matrix $A = [a_{ij}]_{n \times n}$ is a Symmetric Matrix if $a_{ij} = a_{ji}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}_{n \times n} \Rightarrow \boxed{A = A^T}$$

Ex: $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix}$

$\Bigg\} \Rightarrow A = A^T.$

⑥ Skew – Symmetric Matrix:

A Matrix $A = [a_{ij}]_{n\times n}$ is said to be a skew-symmetric Matrix if

$\boxed{a_{ij} = -a_{ji}}$ $\forall$ $1 \le i \le n;\ 1 \le j \le n.$

Ex: $A = \begin{bmatrix} \cancel{a_{11}}^{\,0} & a_{12} & a_{13} \\ \cancel{a_{21}}^{\,-a_{12}} & \cancel{a_{22}}^{\,0} & a_{23} \\ \cancel{a_{31}}^{\,-a_{13}} & \cancel{a_{32}}_{\,-a_{23}} & \cancel{a_{33}}^{\,0} \end{bmatrix}_{3\times 3}$ $\Rightarrow A = \begin{bmatrix} 0 & a_{12} & a_{13} \\ -a_{12} & 0 & a_{23} \\ -a_{13} & -a_{23} & 0 \end{bmatrix}$ $\Rightarrow$ $\boxed{A = -A^T}$

For $i=j$; $a_{ii} = -a_{ii} \Rightarrow 2\cdot a_{ii} = 0$

$\Rightarrow \boxed{a_{ii} = 0}$ ★ All the Principal diagonal elements of a skew-symmetric Matrix are zeroes.

→ The sum of all the elements of a skew-symmetric Matrix is $\underline{\text{zero}}$.

$A = a_{ij} \;\forall\; 1 \le i \le n;\ 1 \le j \le n.$

Sum of all the elements of 'A' $= \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} = k.$

$\Rightarrow \sum_{j=1}^{n} \sum_{i=1}^{n} \cancel{a_{ji}}^{\,-a_{ij}} = k. \Rightarrow k = -\underbrace{\sum_{j=1}^{n} \sum_{i=1}^{n} a_{ij}}_{k}$

$\Rightarrow k = -k \Rightarrow k = 0$

Ex: $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$\text{Sum} = (\cancel{a_{11}}^{0} + a_{12} + a_{13}) + (a_{21} + \cancel{a_{22}}^{0} + a_{23}) + (a_{31} + a_{32} + \cancel{a_{33}}^{0})$

Also

$\text{sum} = (\cancel{a_{11}}^{0} + \cancel{a_{21}}^{-a_{12}} + \cancel{a_{31}}^{-a_{13}}) + (\cancel{a_{12}}^{-a_{21}} + \cancel{a_{22}}^{0} + \cancel{a_{32}}^{-a_{23}}) + (\cancel{a_{13}}^{-a_{31}} + \cancel{a_{23}}^{-a_{32}} + \cancel{a_{33}}^{0}) = -(\text{sum})$

→ A Square Matrix can always be decomposed into sum of a Symmetric & a skew-symmetric Matrix.

i.e $A_{n\times n} = \underbrace{B}_{\text{Symmetric}} + \underbrace{C}_{\text{skew-Symmetric}}$

<u>Proof:</u>

$$A = \frac{1}{2}\underbrace{(A+A^T)}_{B} + \frac{1}{2}\underbrace{(A-A^T)}_{C}$$

Let $B = A + A^T$; consider $\underline{B^T} = (A+A^T)^T = (A)^T + (\cancel{A^T})^T$ A

$= A^T + A = A + A^T = \underline{B}$

(∵ Addition is Commutative)

$\Rightarrow B^T = B$

$\Rightarrow$ $\boxed{B = A + A^T \text{ is always symmetric.}}$

$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$

$3A = \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}$ ✓

<u>Note:</u> A Scalar multiplication doesn't change the symmetricity of Matrix.

Let $C = A - A^T$

Consider $C^T = (A - A^T)^T = (A)^T - (A^T)^T = A^T - A = -\underbrace{(A - A^T)}_{C}$

$$\Rightarrow \boxed{C^T = -C}$$

$\Rightarrow$ $C = A - A^T$ is always skew-symmetric.

$\therefore$ $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ $\longrightarrow$ 'A' is decomposed as sum of a symmetric & a skew-symmetric Matrix.

→ If $A = [a_{ij}]_{5\times 6}$ Where $a_{ij} = i + j$ then Sum of all the elements of Matrix 'A' is <u>195</u>.

Sol: Let $A = [a_{ij}]_{m\times n}$ Where $a_{ij} = i + j$

$$A = \begin{bmatrix} 1+1 & 1+2 & 1+3 & \cdots & 1+n \\ 2+1 & 2+2 & 2+3 & \cdots & 2+n \\ 3+1 & 3+2 & 3+3 & \cdots & 3+n \\ \vdots & \vdots & \vdots & & \vdots \\ m+1 & m+2 & m+3 & \cdots & m+n \end{bmatrix} \begin{matrix} \to n(1) + \frac{n(n+1)}{2} \\ \to n(2) + \frac{n(n+1)}{2} \\ \to n(3) + \frac{n(n+1)}{2} \\ \vdots \\ \to n(m) + \frac{n(n+1)}{2} \end{matrix}$$

$$(+) \to n\left\{\frac{m(m+1)}{2}\right\} + m\left\{\frac{n(n+1)}{2}\right\}$$

$$= \frac{mn}{2}(m+n+2)$$

$$\boxed{\therefore \text{Sum of all the elements} = \frac{mn}{2}(m+n+2)}$$

for $m = 5$ & $n = 6$;

$$\text{Sum} = \frac{5\times \cancel{6}^{3}}{\cancel{2}} \times \{5+6+2\}$$

$$= 15 \times \{13\} = 195$$

→ A Matrix $A = [a_{ij}]_{\underset{\sim}{4\times4}}$ Where $a_{ij} = i^5 - j^5 \; \forall \; 1 \le i \le 4;\; 1 \le j \le 4;$

then Sum of all the elements of the Matrix is $\underline{\;0\;}$.

Sol: Given 'A' is a Square Matrix $A_{4\times4}$.

$$a_{ij} = i^5 - j^5 \Rightarrow a_{ji} = j^5 - i^5 = -(i^5 - j^5)$$
$$= -a_{ij}$$

$$\Rightarrow \boxed{a_{ji} = -a_{ij}}$$

⇒ 'A' is a skew-symmetric Matrix ⇒ Sum of all elements $= 0$.

2 mins Summary

→ Introduction to Linear Algebra.

→ Types of Matrices.

THANK - YOU